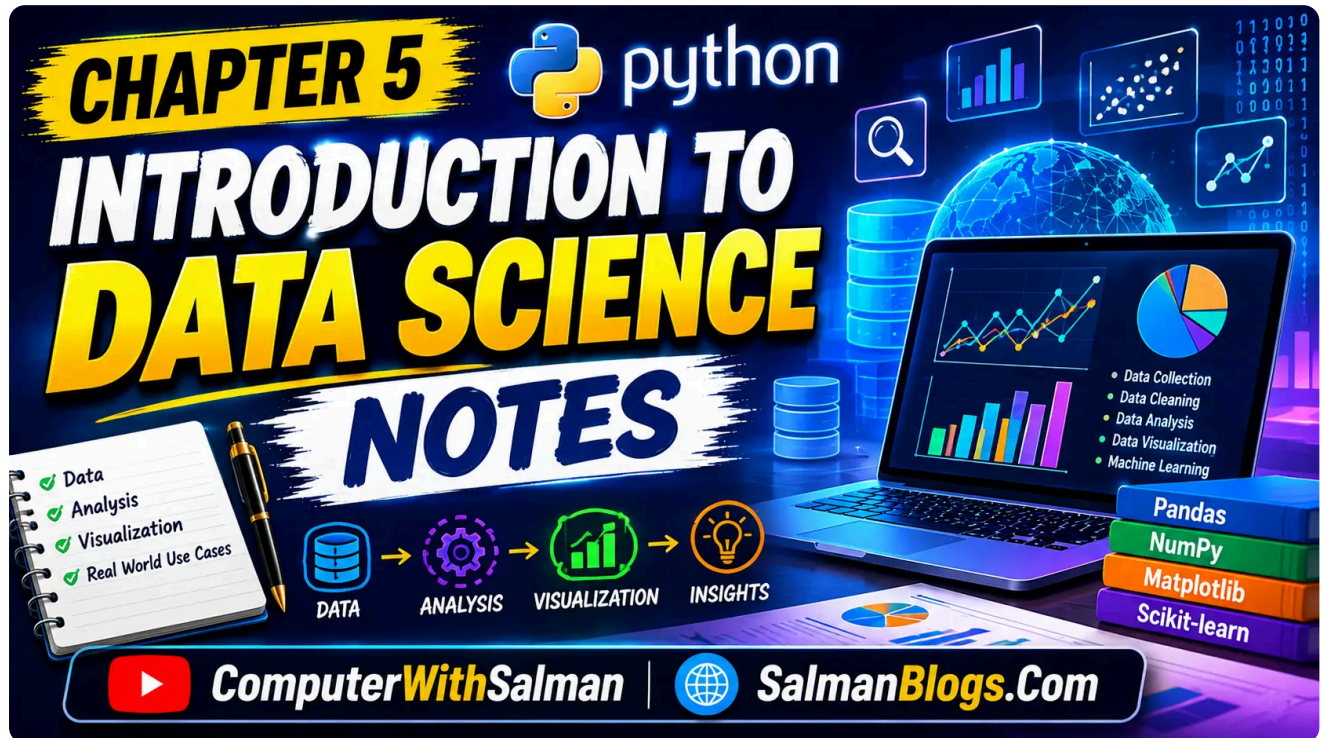


# Introduction To Data Science

Posted in 10 Computer Science



These lecture notes are prepared by **Salman Ahmad**, who is both a **teacher** and a **web developer**. The purpose of this material is to make the topic clear and useful for students of Class 10 Computer Science.

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## Data Science

**Data Science** is the process of **collecting, studying, and understanding data** to find useful information and make better decisions.

### Example

A school can study students' **marks and attendance** to find out which students need more help.

**Easy meaning:**

**Data Science** = Collect data → Study data → Find useful information → Make decisions.

## Importance of Data in Decision Making

**Data** is important in decision making because it provides **useful information and facts** that help people make better decisions.

### Importance of Data

- **Better Decisions**
- **Identify Problems**
- **Find Trends**
- **Improve Performance**
- **Reduce Risk**

## Real-Life Applications of Data Science

### 1. Social Media

Data science is used to understand user interests and recommend content.

**Example:** YouTube recommends videos based on videos you have watched.

### 2. Online Shopping

Data science is used to recommend products based on user activity.

**Example:** An online store recommends products similar to the ones you searched for.

### 3. Healthcare

Data science is used to analyze patient information and help detect diseases.

**Example:** Doctors can analyze patient data to identify possible health problems.

### 4. Transport

Data science is used to analyze traffic and find better routes.

**Example:** Google Maps analyzes traffic data to suggest a faster route.

### 5. Education

Data science is used to analyze students' performance and identify areas for improvement.

**Example:** A school analyzes students' marks to identify students who need extra help.

# Data Science Life Cycle

The **Data Science Life Cycle** is a series of steps used to collect, clean, analyze, understand, and present data to solve a problem.

## 1. Understanding the Problem

The first step is to clearly understand **what problem needs to be solved**.

### Identifying and Defining a Problem

Identify the problem and describe it clearly.

**Example:** A school wants to find out why students' marks are decreasing.

### Setting Clear Objectives

Decide what you want to achieve from the data.

**Example:** Find the main reasons for low marks and identify students who need extra help.

## 2. Collecting Data

Data collection means **gathering the required data** from different sources.

### Methods of Collecting Data

- **Surveys:** Collect information directly from people using questions.  
**Example:** Asking students about their study habits.
- **APIs and Online Sources:** Collect data from websites or online services through **APIs (Application Programming Interfaces)**. An API allows a program to request specific data from an online service and receive it in a usable format.  
**Example:** A weather application can use a **weather API** to request information such as temperature, humidity, and rainfall from an online weather service. The application then displays this data to the user.
- **IoT and Sensor Data:** Collect data from connected devices and sensors.  
**Example:** A temperature sensor records room temperature.
- **Web Scraping:** Collect information from webpages using software or scripts.  
**Example:** Collecting product prices from an online store.

### Choosing the Right Data Source

Not all data is useful. We should choose **reliable, accurate, and relevant** data sources. When selecting a data source, ask:

1. **Is the data from a trusted source?**
2. **Is the data recent or outdated?**
3. **Does the data match the problem I want to solve?**

**Example from book:** If we want to know the **favorite snacks of students in 2025**, we should avoid using a survey conducted in **2020** because students' preferences may have changed.

Other simple and better examples:

- **Weather:** Take data from a trusted weather service, **not an unknown or unreliable website.**
- **Education:** Take data from official school records, **not random online sources.**
- **Business:** Take data from company sales records, **not unverified data from social media.**

## Assessing Data Quality and Bias

**Data quality** means checking whether the data is:

1. **Complete:** No important values are missing.
2. **Correct:** The data does not contain mistakes.
3. **Relevant:** The data matches the problem we are trying to solve.

**Bias** means the data does not represent the full picture and may unfairly favor one group or viewpoint.

**Example:** If we only survey boys about their favorite sports, we will miss the opinions of girls. The results may therefore be **biased toward boys.**

**Remember the following to avoid bias:**

- Ask **different types of people.**
- Keep questions **neutral and not leading.**
- Review the data for **unusual or strange patterns.**

## 3. Cleaning and Validating Data

### Raw Data

**Raw data** is the original data collected before it is cleaned or processed.

**Example:** A student marks file containing missing marks or incorrect entries.

### Common Data Errors

- **Missing Data:** Some values are not available.
- **Duplicate Data:** The same record appears more than once.
- **Outliers:** Values that are unusually high or low compared with other values.

## Missing Data

Some values are not available or are left empty.

| Student | Marks |
|---------|-------|
| Ali     | 85    |
| Ahmed   | —     |
| Sara    | 92    |
| Hamza   | 78    |

## Solution for Missing Data

If a value is missing, we can use **mean, median, or mode** to estimate a suitable value.

| Student | Marks |
|---------|-------|
| Ali     | 80    |
| Ahmed   | —     |
| Sara    | 90    |
| Hamza   | 70    |

For example, using the **mean (average)**:

$$\text{Mean} = (80 + 90 + 70) \div 3 = 80$$

So, we can fill Ahmed's missing marks with **80**.

**Note:** Mean, median, and mode should only be used when they are appropriate for the data.

## Duplicate Data

The same record appears more than once.

| Student | Marks |
|---------|-------|
| Ali     | 85    |

|            |           |
|------------|-----------|
| Sara       | 92        |
| <b>Ali</b> | <b>85</b> |
| Hamza      | 78        |

**Solution:** Identify and remove the duplicate record.

| Student | Marks |
|---------|-------|
| Ali     | 85    |
| Sara    | 92    |
| Hamza   | 78    |

## Outliers

A value is unusually high or low compared with other values.

| Student | Marks    |
|---------|----------|
| Ali     | 82       |
| Sara    | 88       |
| Ahmed   | <b>5</b> |
| Hamza   | 85       |

**Solution:** Check the original source to see if the value is correct. If it is a mistake, correct or remove it.

| Student | Marks     |
|---------|-----------|
| Ali     | 82        |
| Sara    | 88        |
| Ahmed   | <b>85</b> |
| Hamza   | 85        |

**Note:** An outlier is not always an error. It should be checked before changing or removing it.

## Data Validation Techniques

**Data validation** means checking whether the collected data is **correct, valid, and follows the required rules**.

- **Follows correct rules:** Only allowed values are entered.  
**Example:** If the available sports are **Cricket, Football, and Hockey**, "Tennis" should not be entered.
- **Has the right type of values:** The correct type of data is entered.  
**Example:** **Age** should contain numbers, not words.
- **Is not fake or silly:** The data should be **realistic and make sense**.  
**Example:** A student's marks cannot be **150 out of 100**.

## Data Cleaning

**Data cleaning** is the process of finding and **correcting or removing incorrect, incomplete, duplicate, or unnecessary data**.

## Data Cleaning Methods

- **Fill or remove missing values.**
- **Remove duplicate records.**
- **Correct incorrect values.**
- **Handle unusual or incorrect outliers.**
- **Use a consistent format.**

## Why Cleaning Data Matters

Clean data helps us get **accurate analysis and reliable results** by removing errors, missing values, and duplicate records.

**Example:** If a school's student records contain incorrect marks, the analysis of student performance may give **wrong results**.

# 4. Analyzing Data

## What Does It Mean to Analyze Data?

**Data analysis** means **studying data to find useful information, patterns, relationships, and trends**.

**Example:** Comparing students' attendance and marks to see whether attendance affects performance.

## Simple Tools for Data Analysis

Data can be analyzed using simple tools such as:

- **Google Sheets**
- **Microsoft Excel**
- **Manual calculations and tables in a notebook**

## 5. Interpreting Data

**Data interpretation** means understanding what the results of data analysis **actually mean**.

**Example:** After analyzing attendance and marks, we may find that students with higher attendance generally have better marks.

## 6. Visualizing and Communicating Results

After analyzing and interpreting data, the results are presented in a way that is **easy to understand**.

### Importance of Data Visualization

**Data visualization** means presenting data using **charts, graphs, and other visual forms**.

It helps to:

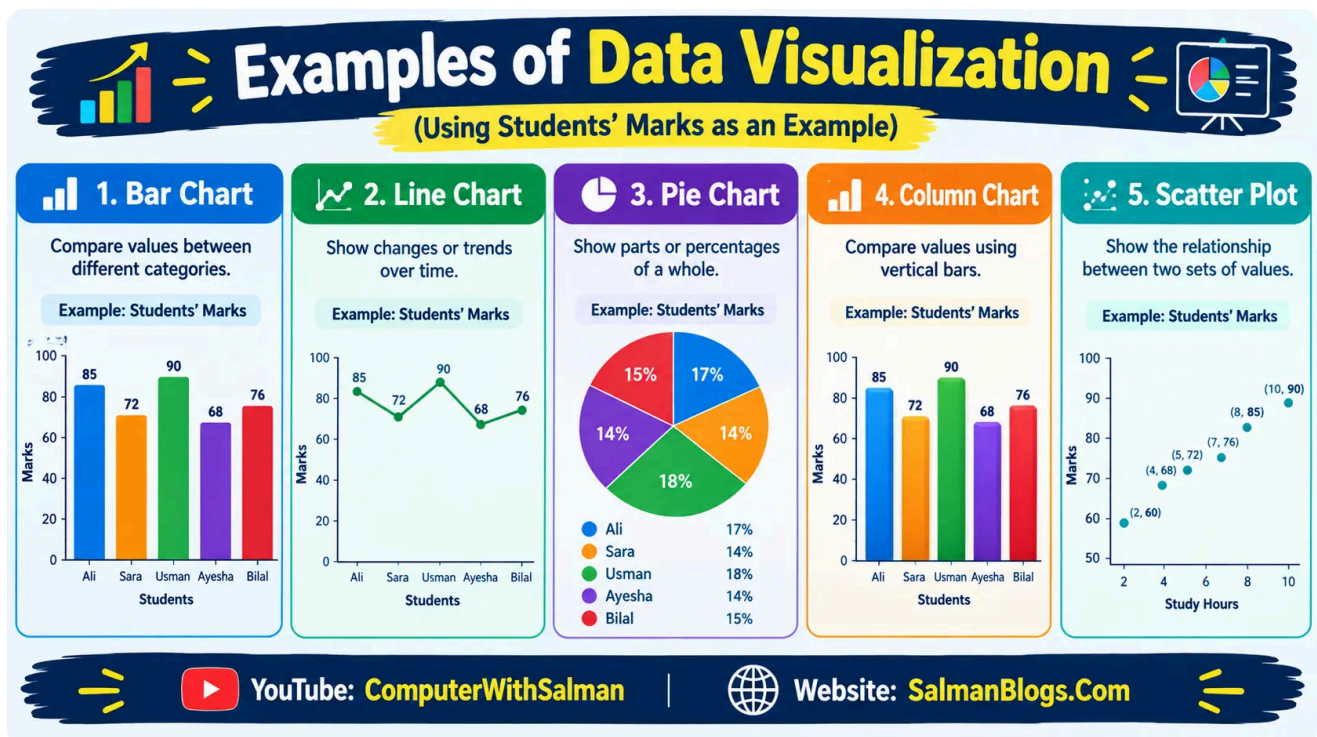
- **Understand data easily.**
- **Identify patterns and trends.**
- **Compare information.**
- **Communicate results clearly.**

### Tools for Visualization

- **Google Sheets**
- **Microsoft Excel**

### Examples of Data Visualization

1. **Bar Chart:** Compare values between different categories.
2. **Line Chart:** Show changes or trends over time.
3. **Pie Chart:** Show parts or percentages of a whole.
4. **Column Chart:** Compare values using vertical bars.
5. **Scatter Plot:** Show the relationship between two sets of values.



## Databases and Storing Data

### What is a Database?

A **database** is an organized collection of data that can be **stored, managed, and accessed easily**.

**Example:** A school's database can store students' names, classes, marks, and attendance.

### Table

A **table** stores data in **rows and columns**.

**Example:**

| ID | Name  | Class |
|----|-------|-------|
| 1  | Ali   | 9     |
| 2  | Ahmed | 9     |

### Record

A **record** is a complete row of data about one item or person.

**Example:** 1 | Ali | 9 is one student record.

### Field

A **field** is a single type of information stored in a column.

**Example:** Name, Class, and ID are fields.

**Note:** In a database table, a **column** is also called a **field**.

## DBMS

**DBMS (Database Management System)** is software used to **create, store, manage, and access data in databases**.

**Examples:** MySQL, Microsoft Access, and PostgreSQL.

### What Can We Do in a DBMS?

A DBMS allows us to perform different operations on data, such as:

1. **Create** databases and tables.
2. **Store** data in an organized way.
3. **Insert** new records.
4. **Update** existing data.
5. **Delete** unwanted data.
6. **Search and retrieve** required data.
7. **Sort and filter** data.
8. **Protect** data using security and access controls.
9. **Create backups** to prevent data loss.

## SQL

**SQL (Structured Query Language)** is a language used to **communicate with and manage databases using commands**. These commands allow us to **create databases and tables, insert, retrieve, update, and delete data**.

**Example:** We can use SQL commands to add a student's record, search for a student's marks, update their information, or delete a record.

### Students Table

Suppose we have the following `Students` table:

| ID | Name  | Class |
|----|-------|-------|
| 1  | Ahmed | 9     |
| 2  | Sara  | 9     |

| ID | Name  | Class |
|----|-------|-------|
| 3  | Hamza | 10    |

## SELECT

The SELECT command is used to **retrieve data** from a database.

```
SELECT * FROM Students;
```

This command retrieves **all fields/columns** and all records from the `Students` table.

| ID | Name  | Class |
|----|-------|-------|
| 1  | Ahmed | 9     |
| 2  | Sara  | 9     |
| 3  | Hamza | 10    |

Here, `*` means **all fields/columns**.

## INSERT

The INSERT command is used to **add new data** to a table.

```
INSERT INTO Students (Name, Class)
VALUES ('Ali', 9);
```

Before using the INSERT command, the table contains:

| ID | Name  | Class |
|----|-------|-------|
| 1  | Ahmed | 9     |
| 2  | Sara  | 9     |
| 3  | Hamza | 10    |

After running the INSERT command, a new record for **Ali** is added:

| ID | Name  | Class |
|----|-------|-------|
| 1  | Ahmed | 9     |
| 2  | Sara  | 9     |
| 3  | Hamza | 10    |
| 4  | Ali   | 9     |

## Easy to Remember

| Command | Action             |
|---------|--------------------|
| SELECT  | Get data           |
| INSERT  | Add data           |
| *       | All fields/columns |

## Exercise – Short Questions

### 1. What is the importance of understanding a problem before collecting data?

Understanding the problem helps us know **what data to collect** and ensures that the data is **relevant and useful**.

**Example:** If we want to find students' favorite sports, we should collect data about their **sports preferences**.

### 2. Name any two methods of collecting data and give one example of each.

**See Topic Methods of Collecting Data.**

### 3. What do we mean by "cleaning" data?

Data cleaning means **finding and correcting or removing errors**, such as missing, duplicate, or incorrect data.

**Example:** Removing a student's duplicate record from a student database.

### 4. Why is it important to validate data before using it?

Data validation ensures that the data **follows the correct rules, has the right type of values, and is realistic**.

**Example:** A student's marks should be a number between **0 and 100**.

### 5. What is the purpose of visualizing data using charts?

Data visualization helps us **understand, compare, and identify patterns or trends** in data more easily.

**Example:** A bar chart can quickly show which student has the **highest marks**.

## 6. How does bias in data affect results?

Bias can make the results **unfair or inaccurate** because the data may not represent the full picture.

**Example:** If we ask only boys about their favorite sports, the results may not represent the opinions of girls.

## 7. Write two steps you will follow to clean a messy survey.

1. Find and handle **missing or incorrect values**.

**Example:** Find a missing student's age from the original survey.

2. Find and remove **duplicate records**.

**Example:** Remove a student's record if it appears twice.

## 8. Name any two tools used for creating charts.

Two tools used for creating charts are **Microsoft Excel** and **Google Sheets**.

**Example:** We can use Excel to create a **bar chart of students' marks**.

**Note:** Dear Students, please visit the website and read the content below if you want to learn how to implement the **Data Science Life Cycle** in an easy way.

# Example: Data Science Life Cycle – Calculating Students' Marks

Suppose a teacher wants to **calculate and analyze the average marks of students** in a class. We can use the Data Science Life Cycle to solve this problem.

## 1. Understanding the Problem

First, we clearly understand what we want to find out. Here, our goal is to **calculate the average marks of students** and understand their performance.

**Example:** We want to find the average marks of 5 students in a Computer Science test.

## 2. Collecting Data

We collect the required data, such as the **names of students and their marks**.

| Student | Marks |
|---------|-------|
| Ali     | 80    |
| Ahmed   | 75    |